

PARALLAX | INVESTIGATION DOSSIER

Investigative lead only; no ownership attribution or legal certification

Address and candidate entity

sim_actor-1_wallet
entity:41b8043fe129120f
Exported: 2026-09-25T03:58:21.043897+00:00

Priority	Evidence completeness	Transactions
83.51/100	100.0/100	12

Reasons for review

High fan-out; Rapid repeated spending; Repeated equal outputs; Address reuse

Cluster scope

1 candidate members; 13 unique supporting transactions. Ownership remains unverified.

sim_actor-1_wallet

Evidence-derived case note

Address sim_actor-1_wallet has 12 observed spending transactions. Fracture Index 83.51/100 (High score band). Estimated decayed seed exposure: 59.4%; supporting path hop counts [1]; references: synthetic actor ground truth / actor-1. This is proportional address accounting, not exact UTXO tracing. Observed patterns: High fan-out, Rapid repeated spending, Repeated equal outputs, Address reuse. No supported cash-out candidate found within the bounded downstream search. Relay associations and pattern matches do not establish ownership or criminality.

Supporting evidence subgraph

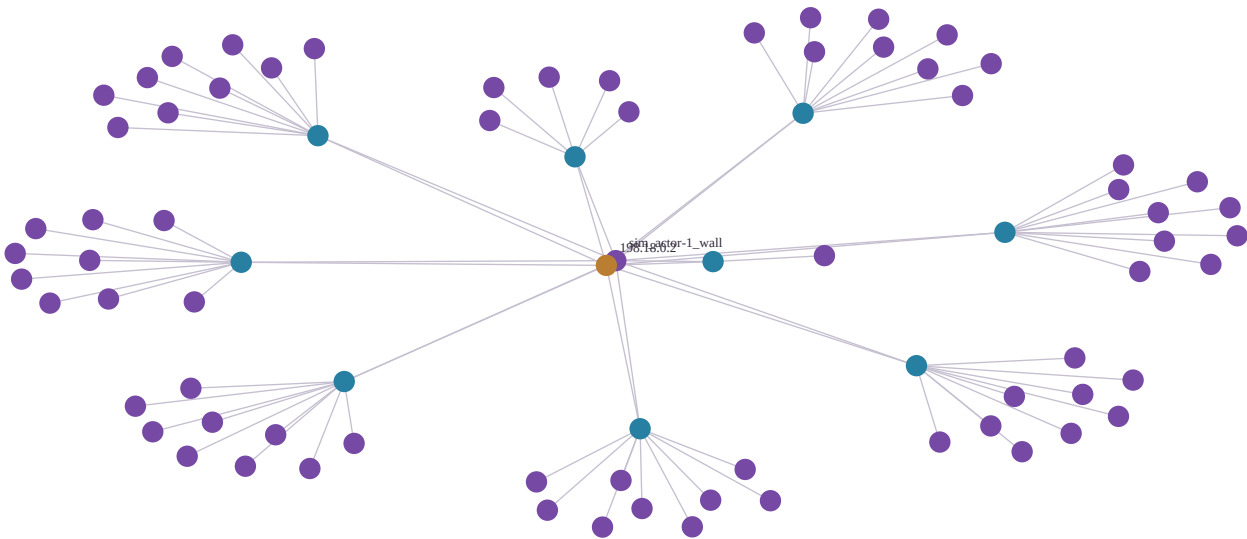


Figure limited to 80/124 nodes. Full graph is in signed case.json.

Purple: address; blue: transaction; amber: observed relay. Labels abbreviated in figure. Full identifiers and evidence in signed case.json.

High fan-out

At least one transaction has 10 outputs; batching is a benign alternative.

Rapid repeated spending

12 transactions with median observed gap 12s; automated services can behave similarly.

Repeated equal outputs

At least half of transactions have 3+ equal outputs; collaborative transactions are not proof of illicit activity.

Address reuse

An output address appeared earlier in this dataset; reuse alone is not illicit.

Model explanation

Tree SHAP values explain Isolation Forest mean path length. Negative values increase anomalousness. These are not causal claims.

equal_output_fraction: value 1.0; SHAP path-length contribution -0.96808

log_mean_btc: value 0.468709; SHAP path-length contribution -0.736847

log_median_gap_seconds: value 2.564949; SHAP path-length contribution -0.610827

log_total_btc: value 2.101099; SHAP path-length contribution -0.549296

fee_rate_cv: value 0.0; SHAP path-length contribution -0.513854

interval_entropy: value -0.0; SHAP path-length contribution -0.467367

Uncertainty and alternative explanations

Network endpoints may be relays, shared infrastructure, or NAT; no ownership attribution.

Anomaly scores measure unusual behavior, not criminality.

Country/ASN enrichment is partly or wholly unavailable; no geolocation is inferred.

Entity membership is a revisable heuristic; change and relay candidates are not merged.

Integrity and reproducibility

Model SHA-256: 71afdbbee4b90b58e769b2f178a9a3d9a2eabe0d941a27d19e587a9b04f82cee0

Audit head: dd3370824000a298b7287139fd718b9850cfa35299db976a3e563b78691d6ade

Verify report.pdf.sig against the exact PDF bytes using the separately trusted public key. The ZIP manifest also covers this PDF. This is a detached Ed25519 signature, not a PDF viewer certificate.

Source references

round-0.jsonl | dadc83843d9531362a9b59cbaea24ba5401816c9765a4f737addc59155dd32a4

Selected supporting observations

2025-10-02T03:49:05+00:00 | source row 26 | TXID 1642deee02f4790564b6cd7bab3430ae21148baf9984b3d806885c47e5ce0cc9 | relay 198.18.0.2 -> 203.0.113.1

2025-10-02T03:47:29+00:00 | source row 18 | TXID 1bd85c0b9ab7f4edb91955ecba0809a8c996e99f7a0c5a72b261b1192865f8ee | relay 198.18.0.2 -> 203.0.113.1

2025-10-02T03:46:40+00:00 | source row 14 | TXID 292e14e9be61f4a6eddf8cf4e3c495c822f8d17cf72d767772b38fa66aacbf2 | relay 198.18.0.2 -> 203.0.113.1

2025-10-02T03:48:05+00:00 | source row 21 | TXID 390265558f18947fb95c479676f9dc1016378d60574ac2c60e2b48e3f59d3765 | relay 198.18.0.2 -> 203.0.113.1

2025-10-02T03:47:17+00:00 | source row 17 | TXID 6c1684813d2ac8a4e2902f84b6e168bf7450ae3a275a63774a56a6ae48213b3d | relay 198.18.0.2 -> 203.0.113.1

2025-10-02T03:47:41+00:00 | source row 19 | TXID 7db34fe59a6b84a730b54c78c9d3d0ac035cfe2d3ccd7b3368c8c50957286bf8 | relay
198.18.0.2 -> 203.0.113.1

2025-10-02T03:47:53+00:00 | source row 20 | TXID 80ab1b46f7c5e3f4239f759656655e144657f89f32630bfb2a09c57659895cda | relay
198.18.0.2 -> 203.0.113.1

2025-10-02T03:47:05+00:00 | source row 16 | TXID 80d318c489e51e17d7a69093d295338c10fd2ac758e380ba584075af2671df89 | relay
198.18.0.2 -> 203.0.113.1

2025-10-02T03:48:17+00:00 | source row 22 | TXID 8958edafc0f1d2939ff8bb389591d6c2c178963063cde7fc1cc9bf5e46c2f8d8 | relay
198.18.0.2 -> 203.0.113.1

2025-10-02T03:48:29+00:00 | source row 23 | TXID 8fab06952417134c286c38d1dd24973cc1aa8fdf28b6e124d2e13d3367df1ee | relay
198.18.0.2 -> 203.0.113.1

2025-10-02T03:48:53+00:00 | source row 25 | TXID b8f38fb2430ef801280d2195f9e5ba20f92109f474c17ee63f89ae8bda34501c | relay
198.18.0.2 -> 203.0.113.1

2025-10-02T03:46:53+00:00 | source row 15 | TXID c312753d083a32d724a4fead6445f1e23d04616c113baf3a985458886ad3c574 | relay
198.18.0.2 -> 203.0.113.1

2025-10-02T03:48:41+00:00 | source row 24 | TXID cf7272b83ad952437b60af6a2d72b1d76d5ddd1056307b5ec8af5a9888365ef5 | relay
198.18.0.2 -> 203.0.113.1